About this PDF

Built only from: **NOC:Computer Networks and Internet Protocol (NPTEL course book, weeks 6, 8)** — https://nptel.ac.in/courses/106105183

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Covers 5 of 6 syllabus sub-topics of this unit.

Partly covered in the source:

- IPV6, ARP, RARP, ICMP, DHCP

Not in the source (not covered in this PDF):

- Switching : Packet Switching

Internet protocol

Definition

The Internet Protocol (IP) is the network-layer protocol that provides addressing and forwarding of packets across interconnected networks. The original and most widely used version is IPv4, which uses a 32-bit address to uniquely identify each host, while a newer version, IPv6, employs a 128-bit address space.

Explanation

IP addresses consist of two parts: a network identifier and a host identifier, similar to a postal address that specifies a region and then a specific location within that region. IPv4 was designed when the Internet was intended mainly for fixed desktop computers, so it lacks built-in support for mobility, quality of service, and security; these features had to be added later as separate extensions. Because IPv4’s 32-bit space became insufficient for the exponential growth of devices, IPv6 was introduced with a much larger address space (128 bits) and a globally unique hierarchical addressing scheme that enables more efficient routing.

Example

An example of an IPv4 address mentioned in the lectures is **254.12.35.129**.

Key points

- IP provides a unique numeric identifier for each host on a network, combining network and host components.
- IPv4 uses 32-bit addresses, while IPv6 expands this to 128 bits to accommodate the growing number of devices.
- IPv4 was originally designed without native support for mobility, QoS, or security, leading to added complexity and the need for patches such as Mobile IP.
- IPv6 introduces globally unique hierarchical addressing, which improves routing efficiency and eliminates the address-space exhaustion problem of IPv4.

IPV4

Definition

IPv4 is the original 32-bit addressing scheme used in the Internet, originally designed for fixed desktop computers and lacking built-in support for mobility and security features.

Explanation

The IPv4 addressing structure was created when connectivity was the primary concern, so each device received a unique 32-bit address. Because it was not intended for mobile devices, additional mechanisms such as Mobile IP had to be added later as patches to provide mobility support. Security, authentication, and auditing were considered secondary requirements and were also added on top of the basic IPv4 design.

Example

A typical IPv4 address such as **202.141.81.2** is used in DNS mapping to reach a NAT box that forwards requests to internal web servers, illustrating how IPv4 works with network address translation to enable access to private networks from the public Internet.

Key points

- IPv4 uses 32-bit addresses, sufficient for early fixed-desktop environments.
- Mobility and security were not inherent to IPv4 and required later extensions like Mobile IP.
- NAT is commonly employed with IPv4 to map public addresses to multiple private servers, enabling load balancing and external access.

IP Addressing

Definition

IP addressing is a hierarchical scheme that uniquely identifies a host on a network by dividing a 32-bit IPv4 address into a network-address part and a host-address part. Historically this division was fixed in classful addressing (classes A, B, C, D, E).

Explanation

The 32-bit address is split so that the network portion identifies the specific network, while the remaining bits identify individual hosts within that network. For each class, a certain number of bits are reserved for the network and the rest for hosts. The all-zero host bits represent the network address, and the all-one host bits represent the broadcast address; both are not assignable to hosts. Consequently, the count of usable host addresses for a network is 2^(host-bits) − 2, subtracting the network and broadcast addresses.

Example

A Class C network has 8 host bits. The total possible host combinations are 2^8 = 256, but the all-zero (network) and all-one (broadcast) addresses are excluded, leaving 256 − 2 = 254 usable host addresses.

Algorithm / Steps

1. Identify the class of the IPv4 address to determine the number of host bits (e.g., Class A = 24 host bits, Class B = 16 host bits, Class C = 8 host bits).
2. Compute the total host combinations as 2^(host-bits).
3. Subtract 2 to exclude the network address (all 0) and the broadcast address (all 1).
4. The result is the number of valid host addresses for that network.

Key points

- IPv4 uses a 32-bit hierarchical address split into network and host parts.
- Classful addressing defined fixed bit boundaries for network and host portions across five classes.
- All-zero host bits denote the network address; all-one host bits denote the broadcast address, and both are omitted from host allocation.
- The number of usable hosts in a network is calculated as 2^(host-bits) − 2.

Subnetting

Definition

Subnetting is the process of dividing an IP network into smaller logical networks (subnets) by borrowing bits from the host portion of the address; the subnet mask indicates how many bits belong to the network/subnet part versus the host part.

Explanation

When bits are taken from the host field to form subnet identifiers, using a pattern of all 0’s or all 1’s for the subnet field creates problems. An all-zero subnet makes the subnet’s network address identical to the original network’s address, while an all-one subnet makes the subnet’s broadcast address identical to the original network’s broadcast address. Therefore, these extreme patterns are normally avoided.

Example

- Using the 0 bit as the subnet identifier yields the subnet network address 192.168.0.0, which is the same as the original network address.
- Using the 1 bit as the subnet identifier yields a subnet whose broadcast address becomes 192.168.255.255, identical to the original network’s broadcast address.

Algorithm / Steps

1. Determine the number of subnets required.

2. Calculate the minimum number of bits needed to represent that many subnets, avoiding the use of all 0’s and all 1’s patterns. For three subnets, three bits are required because two bits would force the use of the prohibited patterns.

Key points

- Subnetting uses bits from the host portion to create a subnet identifier, defined by the subnet mask.
- All-zero and all-one subnet identifiers are avoided because they duplicate the original network’s address or broadcast address.
- The number of bits borrowed must be sufficient to represent the required subnets without resorting to the prohibited all-zero or all-one patterns.

IPV6, ARP, RARP, ICMP, DHCP

Partly covered in the source.

Definition

Neighbor Discovery in IPv6 is a protocol that replaces ARP in IPv4 and enables a node to identify other hosts and routers on its link by mapping IPv6 addresses to link-layer (MAC) addresses.

Explanation

When a node needs to send a packet to another host, it must know both the destination’s IPv6 address and its MAC address. The Neighbor Discovery protocol provides this mapping. A node sends a Neighbor Solicitation message (an ICMPv6 type 135) that includes the target IPv6 address and its own source link-layer address. Unlike IPv4 ARP, the solicitation is not broadcast; it is sent directly to the solicited node whose information is already known to the source. The solicited node replies with a Neighbor Advertisement message containing the target IPv6 address and the corresponding target link-layer address.

Example

Consider four devices A, B, C and D with IPv6 and MAC addresses. If A wants to send a packet to B, A knows B’s IPv6 address but not B’s MAC address. A sends a Neighbor Solicitation packet to B; B responds with a Neighbor Advertisement that includes B’s MAC address, allowing A to forward the packet to B.

Algorithm / Steps

1. Node A prepares a Neighbor Solicitation (ICMPv6 type 135) containing the target IPv6 address and its own source link-layer address.
2. A sends the solicitation directly to the solicited node (e.g., node C) rather than broadcasting.
3. The solicited node receives the solicitation, determines the MAC address for the target IPv6 address, and generates a Neighbor Advertisement message containing the target IPv6 address and the target link-layer address.
4. The solicited node sends the Neighbor Advertisement back to the original requester (node A).

Key points

- Neighbor Discovery replaces ARP in IPv6, providing IPv6-to-MAC address mapping.
- Solicitation messages are unicast to a known solicited node, not broadcast as in IPv4 ARP.
- The exchange uses ICMPv6 messages: type 135 for Neighbor Solicitation and a corresponding Neighbor Advertisement for the reply.
- The process enables a node to discover the MAC address of a destination host before forwarding packets.

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